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# Beyond the Building: A Probabilistic Framework for GPS Assignment Under Positional Uncertainty

**Hafiyyan Hilmy Fawwaz** · Makoto Chikaraishi
Hiroshima University

full presentation

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MOTIVATION

## where did they *really* stop?

In a city, GPS error is often bigger than the distance between a building, the street in front of it and the open space next to it.

The usual fix is to pick the nearest building. But urban researchers have long shown that daily life also happens outside buildings.

> **Jacobs** *the street*
> The Death and Life of Great American Cities, 1961
>
> **Gehl** *the space between*
> Cities for People, 2010
>
> **Whyte** *the small plaza*
> The Social Life of Small Urban Spaces, 2010

THE GAP

## one answer, *too sure* of itself

Picking one building also hides how unsure the GPS is. For people who are **moving**, map matching already gives probabilities. For people who **stay**, nobody had done this. And nobody had covered all three types: building, street and open space.

### OUR THREE QUESTIONS

**RQ1** How can we build GPS error into a method that gives probabilities?

**RQ2** How often are its answers the same as the survey answers?

**RQ3** Does a more complex model help, or does it fail on new stays?

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HOW WE DID IT

## let the uncertainty *speak*

**1 Find candidates**
Buildings, street pieces and open spaces within 10 m of the stay.

**2 Give four scores**
Each candidate gets the scores below.

**3 Add them up**
Each score has a weight. One number per candidate.

**4 Make probabilities**
A logit model gives one probability per place type.

**Time near** — share of the stay spent close to it
**Distance** — how far it is from the stay center
**Longest stretch** — longest unbroken time at it
**Direction** — movement along the street

The weights are learned from the survey answers. The final model keeps three of these scores and adds two more inputs.

### THE DATA

**Building**
**Z-Map TOWN** building outlines

**Street**
**MAPPLE** street center lines, made as wide as the street

**Open space**
All the ground that is left

### FIELD SURVEY IN HIROSHIMA

People visited planned stops and wrote down the place type, and when they arrived and left. Each person carried three or more phones, so we have more GPS data per stop.

We downsampled the survey GPS points to match MILES, the GPS data we want to apply the model to: 152,602 points became 15,687.

| 11 | 3–5 | 280 | 907 |
|---|---|---|---|
| people, 2 sessions each | phones per person | real stops | stays with a known answer |

building 531 · street 185 · open space 191

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WHAT WE FOUND

## right *7 times in 10*, even on new stays

**71.9%** of new stays get the right place type. "New" means the model did not learn from them.

### WHAT IT GETS RIGHT, AND WRONG

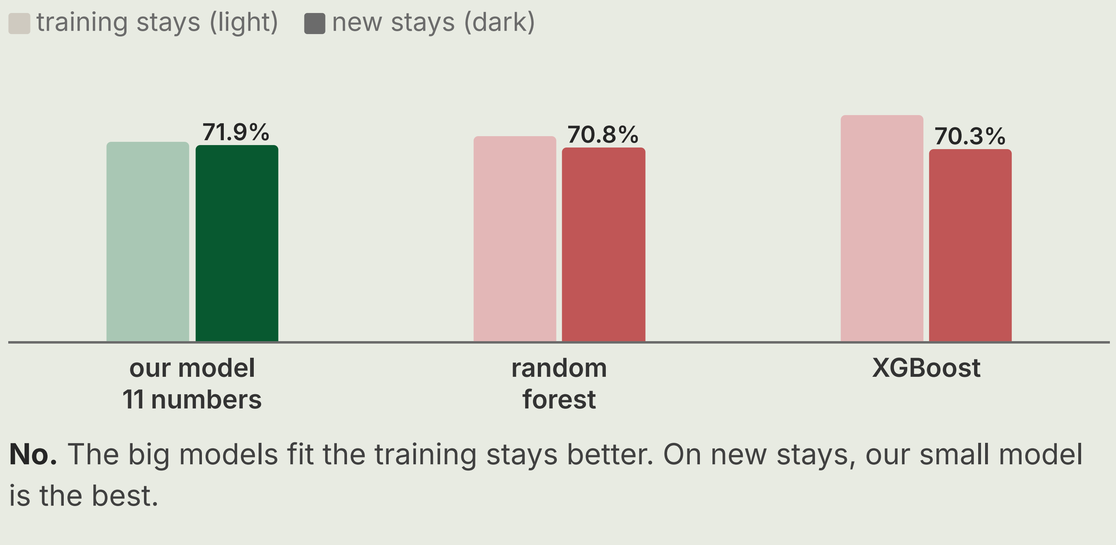
| CORRECT TYPE \ MODEL'S ANSWER | building | street | open space |
|---|---|---|---|
| building | 90.0% (478) | 4.9% (26) | 5.1% (27) |
| street | 41.1% (76) | 48.6% (90) | 10.3% (19) |
| open space | 50.8% (97) | 5.2% (10) | 44.0% (84) |

correct answer · mistake · Small number = stays

**2×**
Buildings are found about **twice** as often as streets or open spaces.

THE TEST

## is a bigger model *better?*




**No.** The big models fit the training stays better. On new stays, our small model is the best.

### WHY ARE STREETS MISSED?

The map cuts one street into many short pieces. A street stay has many street pieces nearby, and the model counts this against the street.

| 13.1 | 7.0 | 4.2 |
|---|---|---|
| street pieces near a street stay | same type near a building stay | same type near an open-space stay |

**New.** A probability for building, street and open space, for people who stay in one place.

**Simple is enough.** 11 numbers beat two machine-learning models on new stays.

**Fixable.** For streets, the weak point is how the map cuts them. It is not the GPS data.

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hafifawwaz@hiroshima-u.ac.jp